CE 310 — Week 12 Assignment

Week: 12 | **Topic:** Sensitivity Analysis **Points:** 60 pts total (3 problems)
Due: Wednesday of Week 13 by 9:00 AM — submit to D2L —

Overview

The lab established sensitivity analysis for the Model 2 and Model 5 regression models — analytical sensitivity, OAT perturbation, tornado diagrams, and annual aggregate effects. This assignment extends sensitivity analysis to new scenarios: a different design temperature, a different model (Model 3 with Season), and a track-specific engineering application.

You will need: your completed lab notebook (Week12_Lab.ipynb), the DOE dataset (Week5_DOE_MedOffice_ZoneTemp_2023.csv), and a Python environment.

How to Submit

> Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → **Run all** — wait until every cell shows a checkmark, no spinners
File format	Download as `.ipynb` — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do **not** edit or delete the `print(f"ANSWER_A1 = ...")` lines
File name	`lastname_firstname_weekN.ipynb` e.g. `smith_jane_week12.ipynb`
Submission	Upload the `.ipynb` file to D2L Dropbox before the deadline

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Problem 1 — Sensitivity Extensions (30 pts)

P1a (8 pts) — Sensitivity at a Different Design Point

In the lab you used T = 85°F (average summer occupied hour) as the design point. Now repeat the tornado diagram analysis at the **1% design temperature: T = 108°F** (the value used in the Week 10 lab for peak cooling predictions).

Perturb each of the three Model 2 parameters (Intercept, Outdoor_Temp_F slope, BH coefficient) by ±10% and compute the swing for each. Report: - The three swing values at T = 108°F - How do the swings compare to those at T = 85°F (from the lab)? - Which swing **increased** the most from T = 85°F to T = 108°F, and why?

Hint: for a linear model, only the slope-related swing depends on T. The BH and intercept swings do not change with T. Verify this numerically and explain it analytically.

P1b (8 pts) — Model 3 Sensitivity: Season Dummies vs. Outdoor Temperature

Model 3 from Week 10 was:

```
m3 = smf.ols('Cooling_kWh ~ Outdoor_Temp_F + C(Season)', data=df).fit()
```

Run the tornado diagram analysis for Model 3 at T = 85°F in Fall season (the reference category, so all Season dummies = 0).

*Dummy variable structure: statsmodels encodes `C(Season)` by dropping the alphabetically first level (Fall) and creating three binary indicators — Spring, Summer, Winter. When Season = Fall, all three dummies are zero and the prediction reduces to Intercept + · T. Each dummy coefficient measures the **incremental** cooling relative to a Fall hour at the same outdoor temperature — not the mean cooling for that season.*

The Model 3 parameters are: Intercept, Outdoor_Temp_F slope, Season_Spring, Season_Summer, Season_Winter. Perturb each by ±10% at the design point (T = 85°F, reference = Fall):

```
i3    = m3.params['Intercept']
b3    = m3.params['Outdoor_Temp_F']
s_spr = m3.params['C(Season)[T.Spring]']
s_sum = m3.params['C(Season)[T.Summer]']
s_win = m3.params['C(Season)[T.Winter]']
base_m3 = i3 + b3*85      # Fall reference (all dummies = 0)
```

Report each swing and rank them in a tornado diagram. Compare: is the outdoor temperature slope still the most sensitive parameter in Model 3? Is the ranking different from Model 2?

P1c (7 pts) — Interaction Model at Peak Design Conditions

Using Model 5 (the interaction model), compute a tornado diagram at T = 103.8°F for the **occupied scenario** (BH = 1).

The four Model 5 parameters are: Intercept, Outdoor_Temp_F, BH, Outdoor_Temp_F:BH. Perturb each by ±10% while holding T = 103.8°F and BH = 1 fixed.

Report the four swing values. Which parameter has the largest swing? Explain physically why the interaction term (Outdoor_Temp_F:BH) has a particularly large swing at high temperatures compared to lower temperatures.

P1d (7 pts) — Sensitivity vs. Uncertainty: Which Matters More?

Consider two sources of prediction uncertainty for the Model 2 annual cooling total:

Source	Description
Temperature measurement bias	Systematic ±5°F error in all outdoor temperature readings
Occupancy schedule change	±260 hours/year change in occupied hours

From the lab: the temperature bias produces 44,936 kWh/yr swing; the schedule change produces 4,642 kWh/yr swing.

(a) Compute the **ratio** of temperature sensitivity to schedule sensitivity.

(b) A building manager can spend $500 to improve the outdoor temperature sensor (reducing its systematic bias from ±5°F to ±1°F) or spend the same $500 to install a more accurate occupancy sensor (reducing schedule uncertainty from ±260 hours to ±52 hours). Using the Model 2 slopes, compute the annual kWh uncertainty reduction from each option. Which investment provides greater reduction in energy prediction uncertainty?

(c) In 2–3 sentences, explain why a linear sensitivity analysis might **overstate** the true impact of temperature bias in this dataset — consider that cooling demand can only be zero or positive (it cannot go below 0 kWh/hr), but the linear model can predict negative values at very low temperatures.

Problem 2 — Track Application (18 pts)

Choose one track. CE students complete Track A. ArcE students complete Track B.

Track A — CE: Chiller Sizing Under Temperature Uncertainty (18 pts)

A mechanical engineer is sizing a chiller using the **Model 5 interaction model**. The ASHRAE 0.4% design temperature is 103.8°F, but the engineer wants to understand how chiller capacity requirements change if the design temperature is uncertain.

(A2-a, 6 pts) Compute the Model 5 occupied-hour prediction at three design temperatures:

T_design	Prediction
98.8°F (−5°F from ASHRAE 0.4%)	?
103.8°F (ASHRAE 0.4% — baseline)	88.55 kWh/hr (from Week 10)
108.8°F (+5°F from ASHRAE 0.4%)	?

Show the calculation for each. Report the swing (max − min prediction) in kWh/hr.

Then convert the swing to **tons of cooling** (1 ton = 3.517 kWh/hr). Report the chiller capacity range in tons that would result from this ±5°F design temperature uncertainty.

(A2-b, 6 pts) Repeat the same analysis using the **additive Model 2** at 98.8°F, 103.8°F, and 108.8°F.

Compare the swings: is the Model 5 swing larger or smaller than the Model 2 swing at this high-temperature range? Explain why the interaction model is more sensitive to design temperature uncertainty than the additive model — specifically, how does the occupied slope (1.78 kWh/°F) vs. the mixed slope (1.03 kWh/°F) explain the difference?

(A2-c, 6 pts) An engineer decides to add a **10% safety factor** to the peak Model 5 prediction to account for design temperature uncertainty.

Compute: what chiller capacity (kWh/hr) does this imply at T = 103.8°F? Convert to tons (1 ton = 3.517 kWh/hr). Is the 10% safety factor sufficient to cover the full ±5°F design temperature uncertainty from A2-a? Show the arithmetic.

Track B — ArcE: Indoor Thermal Comfort Sensitivity (18 pts)

ASHRAE 55-2020 defines acceptable indoor temperature ranges for thermally comfortable office spaces. In a hot-dry climate (cooling mode, summer), the approximate upper comfort limit for 80% acceptability is **76°F** zone temperature in a mechanically cooled building. Zone temperature is driven by outdoor temperature (through the building envelope) and by occupancy (internal heat gains the HVAC system must remove).

Fit a zone temperature regression model:

```
m_zone = smf.ols('Zone_Temp_F ~ Outdoor_Temp_F + BH', data=df).fit()
print(m_zone.summary())
```

(B2-a, 6 pts) Extract the three model coefficients and the R^2. Compute the predicted zone temperature at the design point (T = 85°F, BH = 1).

Build a tornado diagram: perturb each parameter (Intercept, Outdoor_Temp_F, BH) by ±10% while holding T = 85°F and BH = 1, and compute the total swing for each. Report the three swing values and create a horizontal bar chart sorted from largest to smallest.

In 2 sentences, explain why the Intercept produces the largest swing — what does the intercept represent physically in this model, and what does a large intercept swing tell you about HVAC system control vs. outdoor temperature effects?

(B2-b, 6 pts) Solve analytically for the outdoor temperature threshold **T*** at which the zone temperature prediction reaches 76°F during occupied hours (BH = 1):

```
T_star = (76 - intercept - BH_coef) / outdoor_temp_slope
```

Count how many hours in the dataset satisfy `Outdoor_Temp_F > T*` AND `BH == 1` — these are occupied hours predicted to exceed the ASHRAE 55 comfort limit.

Then compute: for a ±5°F systematic outdoor temperature sensor bias, how much does the zone temperature prediction shift at T*? In 2 sentences, explain whether this sensor bias is large enough to cause incorrect ASHRAE 55 compliance classifications, and what that implies for instrumentation accuracy in thermal comfort analysis.

(B2-c, 6 pts) Compare two sensitivity values from this week's work:

Quantity	Value	Interpretation
Cooling energy sensitivity	dCooling/dT = 1.026 kWh/°F (Model 2)	Each 1°F rise requires this much more cooling energy

Quantity	Value	Interpretation
Zone temperature sensitivity	dZone/dT = outdoor temp slope	Each 1°F rise outdoors shifts indoor temperature by this much

Compute the ratio: `energy_per_comfort_degree = (dCooling/dT) / (dZone/dT)`. This tells you how many kWh of additional cooling the HVAC system must consume to suppress 1°F of indoor temperature rise as outdoor temperatures increase.

Report this ratio and explain in 2–3 sentences what it reveals about the HVAC system's performance: is the building's zone temperature well-controlled or highly sensitive to outdoor conditions, and what does the energy cost of that control imply for passive vs. active design strategies?

Problem 3 — Written Reflection (12 pts)

Write a **100–150 word** reflection addressing the following prompt:

> Sensitivity analysis uses the regression coefficients you computed in Weeks 9–10 in a new way — to quantify how much each input drives uncertainty in the output. Describe **one engineering decision** where the tornado diagram ranking from this week's lab (outdoor temperature sensitivity » occupancy sensitivity) would change *what you measure or monitor* in practice. Then describe **one situation** where the OAT assumption (perturbing one input at a time) could give a misleading picture of true uncertainty — and what approach you would use instead.

Grading criteria (12 pts total): - First scenario: specific, plausible, connected to a lab number (4 pts) - Second scenario: identifies a limitation of OAT and explains when perturbing inputs one at a time gives a misleading picture of true uncertainty (4 pts) - Writing quality: concise, within 100–150 words, quantitative (4 pts)

Submission Checklist

- ☐ P1a: three swing values at T=108°F; comparison to T=85°F swings; analytical explanation of which swing changes with T
- ☐ P1b: Model 3 tornado diagram; five swings computed; comparison to Model 2 ranking

- ☐ P1c: four Model 5 swings at T=103.8°F; largest identified with physical explanation
- ☐ P1d: (a) ratio computed; (b) two investment options compared with kWh numbers; (c) limitation of linear model
- ☐ P2 Track A: three temperatures computed; swings compared (M5 vs M2); SF sufficiency shown
- ☐ P2 Track B: zone model fit; tornado diagram (3 swings); T* computed; at-risk hours counted; ratio reported
- ☐ P3: reflection, 100–150 words, two scenarios, quantitative evidence